Decision Radar: Low-Level Design

Version: 0.4 · **Updated:** 21 September 2026 · **Status:** Proposed Workspace-Connected Implementation Contract

1. Architecture & Module Boundaries

Codebase Layout

```
apps/web/                       # Next.js App Router (React + TypeScript)
  src/app/api/v1/[...path]/     # Node.js route handlers forwarding to
FastAPI API
  src/components/               # UI components (cards, evidence viewer,
review forms)
  src/lib/                      # Firebase Auth initialization & API client
services/radar/                 # Python Core Backend (FastAPI + Worker)
  api/                          # FastAPI routes, auth, request validation
  domain/                       # Core entities, state machines, comparison
rules
  connectors/                   # Workspace readers (Docs, Sheets, Meet,
Drive uploads)
  pipeline/                     # Parsing, LLM extraction, comparison,
publication
  adapters/                     # Firestore, Storage, Vertex AI Gemini,
Cloud Tasks
  worker/                       # Private Cloud Tasks & Cloud Scheduler
worker routes
infra/                          # Terraform / GCP deployment specs
```

Module Responsibilities

Module	Primary Contract
`ConnectionService`	Manage user OAuth grants, token refresh, disconnect/reconnect.
`SourceService`	Validate scope, discover native/uploaded Drive sources, persist snapshots.
`SyncService`	Drain Drive change cursors, reconcile recursive folder membership & freshness.
`MeetEventService`	Validate/deduplicate Pub/Sub transcript events, manage subscription renewal.
`SourceAccessService`	Verify viewer native read access for derived alerts, counts, and briefs.
`Normalizer`	Convert Workspace payloads & uploaded file bytes to canonical evidence spans.

Module	Primary Contract
`Extractor`	Call Gemini to extract schema-valid decision candidates and current facts.
`EvidenceValidator`	Validate quote containment and span IDs against stored snapshots.
`DecisionService`	Confirm, edit, and retire decisions with revision control.
`AssessmentService`	Manifest assessment inputs, plan comparison batches, track run coverage.
`Comparator`	Apply deterministic typed rules (date, decimal, enum) to classify evidence.
`ImpactService`	Traverse confirmed dependency graphs to project downstream impacts.
`ReviewService`	Persist human reviewer dispositions and record decision revisions.
`BriefService`	Generate immutable structured brief previews and execute Docs export.
`JobService`	Dispatch transactional outbox tasks, manage worker leases and retries.

Frontend Forwarding & Auth Boundary

- **Same-Origin Proxy:** Next.js proxies browser requests from `/api/v1/*` to the private FastAPI origin.
- **Dual-Header Authentication:**
  - `Authorization: Bearer <user-firebase-id-token>` (user identity verified by FastAPI).
  - `X-Serverless-Authorization: Bearer <gcp-oidc-service-token>` (service identity verified by Cloud Run IAM).
- **Security Protocols:** Static asset caching only; `Cache-Control: private, no-store` for API responses; sensitive headers stripped before forwarding.

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2. Storage Model & Schemas

Firestore Path Structure

Root path: `projects/{projectId}/...` (except global user grants at `workspaceGrants/{grantId}`).

Core Collections Grouped by Subsystem

1. Identity & Scope Subsystem

- `members/{uid}`: Project membership (`admin`, `editor`, `viewer`).
- `connections/{id}`: OAuth grant link, root folder IDs, Meet space names, scope revision.
- `folderIndex/{folderId}`: Ancestry tree, parent IDs, matched root IDs, inventory status.
- `subscriptions/{id}`: Workspace Meet subscription ID, expiration, renewal status.

2. Sources & Storage Subsystem

- `sources/{id}`: Native ID, MIME type, current version ID, status, freshness timestamp.

- `sourceVersions/{id}`: Content hash, GCS object generation, normalizer version.
- `spans/{id}`: Normalized code-point locator `hash(versionId, normalizerVersion, start, end)`.
- `sheetMappings/{id}` / `fileTableMappings/{id}`: Table header, row key, typed column definitions.

3. Decisions & Facts Subsystem

- `decisions/{id}` & `decisionRevisions/{decisionId_rev}`: Statement, owner, rationale, assumptions.
- `assumptions/{id}`: Entity ID, predicate, operator, expected value, unit/currency, criticality.
- `facts/{id}`: Extracted/typed fact, entity mention, predicate, value, publication/effective dates.
- `dependencies/{id}`: Directional graph edge (`enables` / `requires_review_of`).

4. Execution & Audit Subsystem

- `assessmentRuns/{id}`: Register/corpus revisions, run state (`queued`, `completed`, `partial`), coverage.
- `findings/{id}` & `alerts/{id}`: Finding fingerprint, classification, baseline & update evidence.
- `reviews/{id}`: Action (`kept`, `revised`, `retired`, `dismissed`), actor, before/after decision revision.
- `briefs/{id}` & `exports/{id}`: Immutable preview digest, export status, output Google Doc ID.
- `jobs/{id}` & `auditEvents/{id}`: Transactional task outbox, worker lease tokens, audit logs.

Key Data Structures

Assumption Record Example

```
{
  "id": "asm_delivery_01",
  "decisionId": "dec_launch_01",
  "decisionRevision": 1,
  "entityId": "vendor_acme",
  "predicate": "delivery_date",
  "operator": "on_or_before",
  "expectedValue": {"type": "date", "value": "2026-10-10", "precision":
"day"},
  "criticality": "high",
  "origin": "extracted",
  "evidenceRefs": [{"sourceVersionId": "src_v1", "spanId": "span_07",
"quote": "delivery by 10 October 2026"}]
}
```

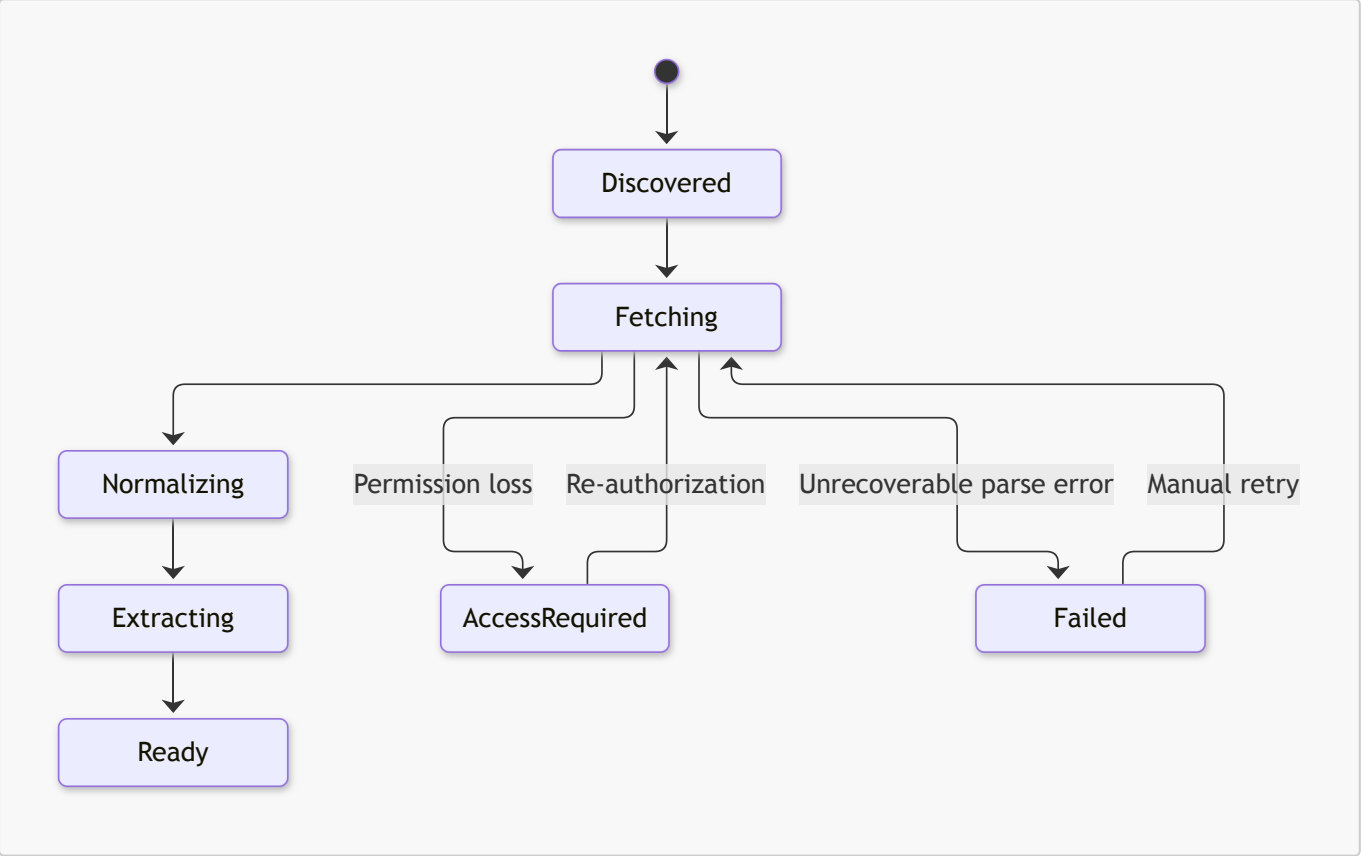
Finding Record Example

```
{
  "id": "fnd_01",
```

```
  "assumptionId": "asm_delivery_01",
  "decisionRevision": 1,
  "classification": "contradicts",
  "reasonCode": "DATE_AFTER_REQUIRED_BY",
  "oldValue": "2026-10-10",
  "newValue": "2026-10-24",
  "delta": {"value": 14, "unit": "calendar_days"},
  "baselineEvidence": [{"sourceVersionId": "src_v1", "spanId": "span_07",
"quote": "delivery by 10 October 2026"}],
  "newEvidence": [{"sourceVersionId": "src_v2", "spanId": "span_02",
"quote": "delivery is now scheduled for 24 October 2026"}]
}
```

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3. State Machines & Invariants



Core Domain Invariants

1. **Confirmed Monitoring:** Only active, confirmed decision revisions are assessed by the comparison engine.
2. **Deterministic Contradiction:** Direct alerts require confirmed baseline evidence, valid update evidence, and a failing typed comparison rule.
3. **No Automatic Mutations:** Model outputs never automatically edit decisions, alter dependency links, resolve alerts, or publish Docs.
4. **Graph Safety:** Dependency graphs enforce directed acyclic structures; self-edges and cycles are rejected at write time.

5. **Re-Review Isolation:** Re-running an assessment preserves existing human review dispositions unless a decision revision or material evidence changes.
6. **Viewer Permission Enforcement:** Derived alerts and evidence display only content for which the viewing user holds active native Workspace read permissions.

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4. Public API Contracts

Base Path: `/api/v1/projects/{projectId}`

API Endpoint Matrix

Method	Endpoint Path	Request Body / Query	Success / Error Codes	Description
`POST`	`/connections`	`{ grantId }`	`201 Created` / `409`	Connect Workspace grant to project.
`PATCH`	`/connections/{id}/scope`	`{ folderIds, fileIds, meetSpaces }`	`200 OK` / `422`	Set monitored Drive folders/files & Meet spaces.
`GET`	`/sources`	`?status=ready&limit=50`	`200 OK`	List monitored sources with health status.
`PUT`	`/sources/{id}/sheet-mapping`	`{ sheetId, range, rowKey, columns }`	`200 OK` / `422`	Set spreadsheet table mapping rules.
`GET`	`/decisions`	`?lifecycle=active`	`200 OK`	List decision cards & baseline evidence.
`POST`	`/decisions`	`{ statement, owner, assumptions }`	`201 Created`	Create a draft decision manually.

Method	Endpoint Path	Request Body / Query	Success / Error Codes	Description
POST	/decisions/{id}/confirm	If-Match: revision	200 OK / 409	Confirm decision draft for active monitoring.
PATCH	/decisions/{id}	{ statement, rationale, assumptions }	200 OK / 409	Update decision (creates new revision).
POST	/assessments	{ reason }	202 Accepted	Trigger on-demand assessment run.
GET	/alerts	?reviewState=open	200 OK	Paginated review alerts for published run.
POST	/alerts/{id}/reviews	{ action, rationale, decisionRevision }	200 OK / 409	Submit human review (Keep, Revise, Retire).
POST	/briefs	{ runId, alertIds }	201 Created	Generate immutable brief preview.
POST	/briefs/{id}/exports	{ provider: "google_docs" }	202 Accepted	Export preview brief to Google Docs.

Protocol Requirements

- **Idempotency:** All state-changing endpoints accept `Idempotency-Key: <uuid>`. Duplicate requests within 24 hours return cached responses.
- **Optimistic Locking:** Mutations require `If-Match: "<revision>"`. Mismatched revisions return `409 Conflict`.
- **Standardized Error Payload:**

```
{
  "error": {
    "code": "REVISION_CONFLICT",
    "message": "Decision updated by another user. Reload required.",
    "requestId": "req_88491",
    "retryable": false
  }
}
```

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5. Workspace Sync & Normalization Engine

5.1 OAuth & Scope Requirements

Scope	Usage Boundary
`https://www.googleapis.com/auth/drive.readonly`	Read-only discovery & ingestion of allowed Drive subtrees and uploaded files.
`https://www.googleapis.com/auth/meetings.space.readonly`	Read-only transcript retrieval for configured Meet spaces.
`https://www.googleapis.com/auth/drive.file`	Incremental scope requested strictly during user-initiated Google Doc brief export.

5.2 Drive Ingestion Algorithm

1. **Start Token:** Call `changes.getStartPageToken` prior to initial folder enumeration.
2. **BFS Inventory Walk:** Breadth-first crawl of specified root folder IDs. Enqueue discovered subfolders up to 250 folders / 100 file limit.
3. **Cursor Drain (1 min):** Poll `changes.list` every minute. If ancestor check confirms file is within allowed tree, queue fetch job.
4. **Periodic Reconcile (5 min):** Re-verify tree ancestry, detect moved/deleted subfolders, poll Sheet ranges, check recent Meet transcript artifacts.

5.3 Ingestion Adapters & Limits

Adapter	Locators & Spans	Limits & Boundaries
Google Docs	Tab ID, heading path, paragraph index, table cell coordinates.	Capture all tabs; ignore embedded drawings/comments.
Google Sheets	Sheet ID, A1 cell range, row business key.	Formula text captured; cell effective value used for comparison.

Adapter	Locators & Spans	Limits & Boundaries
Uploaded Files	Text PDF page/block; DOCX paragraph; CSV/XLSX row keys.	≤ 10 MB, ≤ 50 PDF pages, ≤ 100k chars. Macro execution blocked.
Google Meet	Conference name, transcript entry, speaker, timestamp span.	7-day initial lookback; 30-day Meet API artifact availability window.

6. Comparison Logic & Model Contracts

Schema-Constrained Gemini Extraction

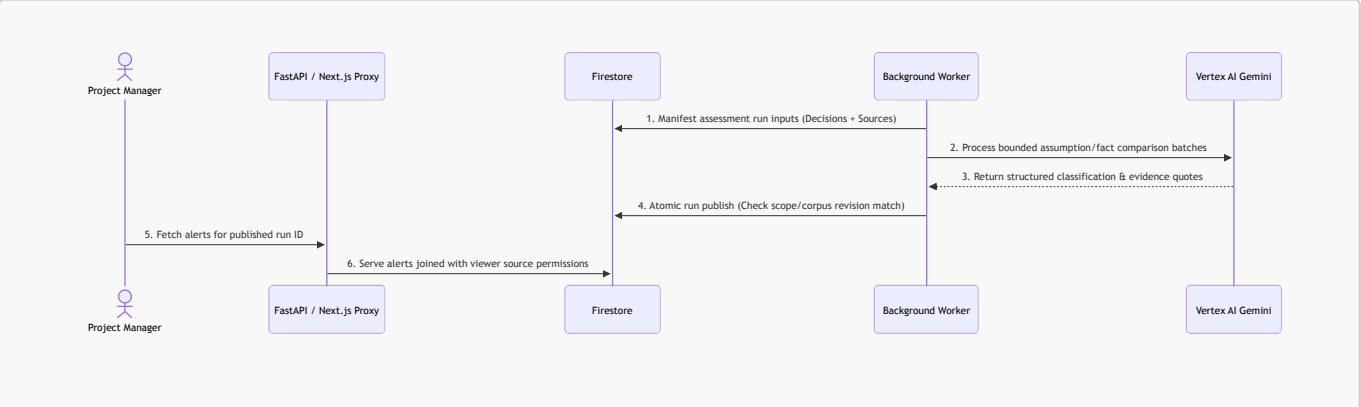
Gemini processes source text spans using structured JSON output mode to extract:

1. **Decision Candidates:** Suggested statement, owner, rationale, typed assumptions.
2. **Current Facts:** Extracted claims, entity mentions, predicate, typed value (date, amount, status), modality (`confirmed`, `tentative`, `negated`).

Typed Comparison Rules Matrix

Target Predicate	Comparison Rule	Result Classification
Date	Updated date strictly after assumption `on_or_before` date	`contradicts` (Δ calendar days)
Numeric / Budget	Updated decimal value strictly exceeds assumption max bound	`contradicts` (Δ value)
Status / Enum	Update confirms cancellation/failure of prerequisite	`contradicts`
Value Match	Update satisfies expected bound	`supports`
Ambiguous / Unlinked	Unapproved alias, missing reference date, tentative wording	`uncertain`

7. Assessment Publication & Concurrency



8. Task Delivery & Retry Policy

- **Outbox Pattern:** Task outbox records created in the same Firestore transaction as state changes.
- **Worker Leases:** Worker locks job with a random lease token (6-minute expiry). Stage timeout set to 4 minutes.
- **Retry Backoff:** Exponential backoff with jitter up to 3 attempts for transient 5xx/429 errors. Terminal failures (e.g., schema drift) pause job and surface UI warnings.

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9. Brief Generation & Google Docs Export

1. **Snapshot Preview:** Immutable brief created from completed assessment run & human review records.
2. **Explicit User Export:** PM clicks export, granting incremental `drive.file` scope.
3. **Doc Creation:** Worker creates Google Doc via Drive API, writes content via Docs API, and records the output document ID.
4. **Idempotent Recovery:** Export key stored in `appProperties` prevents duplicate document creation on retries.

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10. UI Navigation & Component Matrix

Screen Path	Key Components	Active State / Polling
`/projects/[id]`	Overview metrics, health summary, stale run warnings.	Polling source status every 10s.
`/projects/[id]/connections`	Folder picker, Meet space setup, pause/disconnect actions.	Immediate setup validation.
`/projects/[id]/sources`	Folder tree view, upload inventory, Sheet range mapper.	Progress bar during inventory crawl.
`/projects/[id]/decisions`	Decision cards (Draft/Active/Retired), assumption editor.	Instant optimistic UI updates.
`/projects/[id]/reviews`	Alert side-by-side evidence viewer, impact graph, review form.	Revision checks on submission.
`/projects/[id]/briefs/[briefId]`	Formatted preview, evidence citation links, Doc export button.	Async export status tracking.

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11. Security & Operational Rules

- **Viewer Access Filter:** Server checks `requiredSourceIds` against viewer's Workspace grant before returning any derived alert, count, or brief.
- **Credential Storage:** Refresh tokens encrypted via Cloud KMS; client secrets stored in Secret Manager.
- **Log Privacy:** Structured logging strips source text, user OAuth tokens, document titles, and prompt contents.

12. Validation Plan & Acceptance Criteria

Acceptance Gates (1/2)

- **Typed Comparisons:** Correctly identify date delays and budget overruns across leap years and timezone boundaries.
- **Provenance Integrity:** 100% of published quotes match exact stored code-point locators in GCS snapshots.
- **Idempotency:** Re-running sync or task retries produces zero duplicate alerts or Firestore records.

Acceptance Gates (2/2)

- **Drive Discovery:** Recursive subfolder discovery and file change cursor drains execute cleanly without missing files.
- **Access Control:** Revoking user read permission on a Workspace file immediately hides associated derived alerts for that user.
- **End-to-End Verification:** Edit in Doc/Sheet or new Meet transcript automatically generates a cited alert ready for PM review within target latency bounds.

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13. Implementation Sequence

1. **Phase 1 (Days 1–2):** Project scaffold, Firebase auth proxy, FastAPI routes, KMS token encryption.
2. **Phase 2 (Days 3–5):** Drive recursive folder crawl, cursor drain pipeline, GCS snapshot storage.
3. **Phase 3 (Days 6–7):** Sheets/file table mappers, Gemini extraction adapter, typed comparison engine.
4. **Phase 4 (Days 8–10):** Workspace Events Meet subscription, Pub/Sub webhook handler, transcript parser.
5. **Phase 5 (Days 11–14):** Impact graph traversal, review workflow state machine, Google Docs exporter.
6. **Phase 6 (Days 15–17):** Held-out synthetic test suite execution, quality metrics validation.
7. **Phase 7 (Days 18–21):** UI polish, operational logging setup, final demo preparation.